(Please write your Enrollment Number) Enrollment No._____

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:<B. Tech. (CSEAI)>, Semester:<5>
(Nov-Dec, 2022)

Subject Code: BAI-305	Subject: Deep Learning
Time :3 Hours	Maximum Marks :60

Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1 (5*4=20)

(a) Differentiate between sigmoid and softmax using formulae.
(b) Design a neural network (two inputs, x1 & x2) which outputs
out = x1 + x2, if (x1+x2) >0 else out = 0
Clearly mention the weights, activation function, number of hidden layers, and number of neurons in each hidden layer.
(c) Explain RNN using all equations (all possible weights), assume input length 2.
(d) Draw a neat diagram of CONV1-POOL1-CONV2-POOL2, choose hyper parameters appropriately, find learnable parameters. Assume an input image of size 10x10x3.

UNIT-I

Q2 Using diagrams explain linear separable data. Give examples of non-linear data (using diagrams), and explain how non-linear data can be pre-processed or transformed (give equations) into linear form. (10)

Q3 Logistic regression is the basic unit for neural networks. Comment on this statement using equations. How is loss computed in logistic regression, explain using an example and relevant equation. (10)

UNIT-II

Q4 Explain hyper parameter tuning using a concrete example (say decision tree). Clearly mention the parameters tuned, their definition, possible values, and intuition. (10)

Q5 Explain at least three loss functions (with equations) with suitable examples. (10)

UNIT-III

Q6 Explain RNN using a suitable diagram for solving Named Entity Recognition in the field of NLP. Write all relevant equations. (10)

Q7 Explain GRU and LSTM with all relevant equations. (10)

UNIT-IV

Q8 Consider a 3x3 image [[3,4,2],[6,5,4],[3,2,1]] and a filter of size 2x2 [[-1,0],[1,0]]. Perform convolution operation in following settings, and give your answers: (10)
(1) pool=0, stride=1
(2) pool=1, stride=2

Q9 Suppose there are two images (each of size 2x2) in the train dataset with values: [[3,4],[2,1]] and [[1,3],[2,4]], their corresponding labels are 0 and 1. Apply KNN (use appropriate distance function) to classify a test image [[1,2],[2,3]]. (10)